

Logic PhD Exam, August 2019

Solve 5 problems of the following; at least one from each section.

A. Set Theory.

1. Sketch the construction of any model of ZFC and the negation of the Continuum Hypothesis.
2. Define the class of Δ_0 formulas. Prove carefully that if $\phi(x)$ is a Δ_0 formula with one free variable and M is any transitive set and $x \in M$, then $\phi(x)$ is equivalent to $M \models \phi(x)$.
3. Prove that if X is a Polish space and $A_0, A_1 \subset X$ are disjoint analytic sets, then there are disjoint Borel sets $B_0, B_1 \subset X$ such that $A_0 \subset B_0$ and $A_1 \subset B_1$.

B. Computability.

1. Define $0'$ (zero jump) and show that it is a computably enumerable set which is not computable.
2. Define the many-one reducibility and sketch the proof that there is a many-one degree strictly between 0 and $0'$.
3. What is Gödel's diagonalization lemma? Prove the lemma.

C. Model theory.

1. State the downward Löwenheim–Skolem theorem and prove it.
2. State the Łoś' theorem and prove it.
3. Find a complete theory with exactly one infinite countable model up to isomorphism. Prove this property of the theory.